THE ZERO PARADOX

ZP-M: Kleene–Ordinal Bridge

Initial: May 2026 | Current: 1.17, November 2026

Synthesis bridge. All theorems §I–§IV proved sorry-free in Lean 4. Closes the hfp free hypothesis (ZP-L) and the snapEmbed type bridge. Axiom footprint: [propext, Classical.choice, Quot.sound] throughout.

ZP-M bridges the two prior layers — ZP-K (computational grounding via Kleene's second recursion theorem) and ZP-L (ordinal snap threshold at ε_0) — with two formal results. First, the free hypothesis hfp in snap_exactly_at_epsilon_zero (ZP-L §VI) is proved to follow from a minimal alignment hypothesis plus monotonicity: hfp is not an independent condition. Second, the canonical type bridge snapEmbed : MachinePhase → $\mathbb{Z}_2$ formalizes the structural triangle connecting the pre-snap state c_0 (read as the Kleene quine, a KleeneStructure requirement in ZP-K, not a theorem), the ordinal snap at ε_0 (ZP-L), and the 2-adic limit 0 (ZP-B).

The closing result (§IV) proves two existence statements together: a code c with eval c n = eval c (encode c + n) for every n (a periodicity condition, proved via the recursion theorem, which constant codes also meet) and the least ordinal fixed point $\varepsilon_0 = \omega^{\varepsilon_0}$. It is a conjunction of two existence proofs over different types. The two domains — computability theory (codes and Gödel numbers) and ordinal theory (ω-tower iteration) — use no shared mathematical machinery; reading them as one pattern is the framework's interpretation, not what the theorem proves (Remark R-M.1). R-M.1 also tracks what this means for DA-1 Path 2.

Section I: The snapEmbed Morphism

snapEmbed is the canonical type bridge from ZP-E's machine-phase state space to ZP-B's 2-adic integers. The pre-snap state c_0 maps to 1 (a 2-adic unit); the snap state c_1 maps to 0 (the 2-adic limit of the tower images). Under multiplication on $\mathbb{Z}_2$, 0 is absorbing — mirroring exactly the join structure on MachinePhase, where c_1 is the absorbing element.

Within the ZP framework, $0 \in \mathbb{Z}_2$ plays the role of $\bot$ — it is the 2-adic valuation limit and the fixed point of multiplication by 2. snapEmbed makes this modelling commitment concrete: c_1 maps to 0 because c_1 IS the object at infinite 2-adic depth. This is a modelling commitment, not a ring-theoretic theorem.

Definition: snapEmbed (Incompleteness.lean §I)
snapEmbed : MachinePhase → $\mathbb{Z}_2$
snapEmbed $c_0 = 1$ (pre-snap: 2-adic unit)
snapEmbed $c_1 = 0$ (snap state: 2-adic zero = 2-adic limit of tower)

Definition: snapEmbed (Incompleteness.lean §I)

The morphism property: join on MachinePhase (c_1 is absorbing under join) corresponds to multiplication on $\mathbb{Z}_2$ (0 is absorbing under $\times$). Both structures share the absorbing-element pattern — the bridge is structural, not incidental.

Theorem: snapEmbed_injective (Incompleteness.lean §I)

Function.Injective snapEmbed

c_0 and c_1 map to distinct 2-adic integers ($1 \neq 0$ in $\mathbb{Z}_2$).

Proof: case split on MachinePhase; simp_all closes each case.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: snapEmbed_mul_morphism (Incompleteness.lean §I)

$\forall$ a b : MachinePhase, snapEmbed (join a b) = snapEmbed a $\times$ snapEmbed b

snapEmbed sends the join operation to multiplication.

Under this map, the absorbing-element structure is preserved:

join c_1 x = c_1 for all x $\leftrightarrow$ 0 $\times$ y = 0 for all y $\in \mathbb{Z}_2$.

Proof: case split; simp [snapEmbed].

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Lemma: snapEmbed_c0_val (Incompleteness.lean §I)

(snapEmbed c_0).valuation = 0

c_0 maps to a 2-adic unit: valuation 0, norm 1.

Proof: simp via PadicInt.valuation_one.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Lemma: snapEmbed_c1_dvd (Incompleteness.lean §I)

$\forall$ n : $\mathbb{N}$, (2 : $\mathbb{Z}_2$)^n | snapEmbed c_1

c_1 maps to 0, which is divisible by all powers of 2.

0 $\in \mathbb{Z}_2$ is divisible by 2^n for every n — infinite 2-adic depth.

This is the formal signature of $\bot = \{\bot\}$: infinite 2-adic valuation.

Proof: simp [snapEmbed_c1].

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Remark: snapEmbed and the bottom role

snapEmbed establishes the morphism property (join $\mapsto \times$) and the absorbing-element correspondence. Relative to the ordinals' floor, the bottom-role occupant 0 and the ε_0 occupant differ (epsilon0_ne_bot); against the 2-adic 0 an identity with ε_0 is not a well-formed statement (it fails to elaborate, Ordinal vs $\mathbb{Z}_2$), and the link runs through maps: snapEmbed sends c_1 to 0, the Scale.lean chart's ($\bot = 0$) bottom-role occupant. The canonical threshold map Ordinal → MachinePhase is order-non-decreasing in ZP-A's join order (snap_map_mono), and no map Ordinal → MachinePhase is compatible with the CNF → $\mathbb{Z}_2$ map along the tower: for every g : MachinePhase → $\mathbb{Z}_2$ the square fails, because MachinePhase has two values and the images of stages 1, 2, 3 have three distinct 2-adic valuations (the example after c1_epsilon_zero_identification in ZeroParadox/Ordinal/Gentzen.lean); through snapEmbed it fails at every stage $n \geq 1$, because snapEmbed takes only the values 1 and 0, and stage $n \geq 1$ maps to 2^n, which is neither (ZeroParadox/Ordinal/Incompleteness.lean § II).

What the bridge achieves: c_1 (the snap state assigned at ε_0) maps to 0 (the 2-adic limit) under snapEmbed. The triangle is formally exhibited. Whether Classical.choice is forced by the ZP metric collapse is a separate open question (ZeroParadox/Ordinal/SyntacticCollapse.md).

Section II: Closing the hfp Gap

snap_exactly_at_epsilon_zero (ZP-L §VI) carries a free hypothesis hfp asserting that for any map ϕ : Ordinal → MachinePhase, every fixed point of $\omega^{\wedge}\cdot$ maps to c_1. This hypothesis is required for the upper-bound direction of the minimality argument (ε_0 is the first snap point). The canonical witness in epsilon_zero_snap_canonical (ZP-L §VII) satisfies hfp directly by case analysis, but hfp is not derived from the structural axioms.

ZP-M §II closes this gap. Given monotonicity and $\phi\ \varepsilon_0 = c_1$ as a minimal alignment hypothesis, hfp follows. The argument: for any fixed point α of $\omega^{\wedge}\cdot$, epsilonZero_le_fixedPoint gives $\varepsilon_0 \leq \alpha$; monotonicity then chains $\phi\ \varepsilon_0 \leq \phi\ \alpha$; substituting c_1 and using c_1's absorbing property gives $\phi\ \alpha = c_1$.

Theorem: hfp_from_epsilon_zero (Incompleteness.lean §II)

For any ϕ : Ordinal → MachinePhase satisfying:

hmono: $\forall\ \alpha \leq \beta$, join ($\phi\ \alpha$) ($\phi\ \beta$) = $\phi\ \beta$

hε_0: ϕ epsilonZero = c_1

we have: $\forall\ \alpha$, $\omega^{\wedge}\alpha = \alpha \to \phi\ \alpha = c_1$.

Proof: epsilonZero_le_fixedPoint gives $\varepsilon_0 \leq \alpha$; hmono gives join c_1 ($\phi\ \alpha$) = $\phi\ \alpha$; join c_1 x = c_1 for all x (absorbing property of c_1), so $c_1 = \phi\ \alpha$.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: snap_unconditional (Incompleteness.lean §II)

For any ϕ satisfying hmono, h0 (tower stages $\mapsto c_0$), and hε_0 ($\phi\ \varepsilon_0 = c_1$):

Theorem: snap_unconditional (Incompleteness.lean §II)
$\phi\ \varepsilon_0 = c_1 \wedge \forall\ \alpha, \phi\ \alpha = c_1 \to \varepsilon_0 \leq \alpha$.
ε_0 is the minimal snap threshold under minimal hypotheses.
Proof: snap_exactly_at_epsilon_zero with hfp supplied by hfp_from_epsilon_zero.
Lean purity: [propext, Classical.choice, Quot.sound]. ✓

What §II achieves: hfp is no longer a free hypothesis in the minimality theorem. Given $\phi\ \varepsilon_0 = c_1$ (the alignment condition: the map snaps at ε_0) plus monotonicity, the full minimality result follows from the ordinal structure alone. hfp is a derived consequence, not an additional commitment.

Section III: The Kleene–Ordinal Triangle

ZP-K established: $c_0 = \bot$ is a Quine atom (da1_closed_concrete : IsQuineAtom ($\bot$: MachinePhase)). ZP-L established: the canonical snap map assigns c_1 exactly at ε_0, with tower images in $\mathbb{Z}_2$ converging to 0. The triangle connects all three objects formally:

Edge	Objects connected	Formal content
A ↔ C (ordinal snap)	$\varepsilon_0 \leftrightarrow c_1$	Canonical snap map: stages below $\varepsilon_0 \mapsto c_0$; $\varepsilon_0 \mapsto c_1$ (ZP-L §VII)
A ↔ B (2-adic convergence)	Tower → $0 \in \mathbb{Z}_2$	cnfToZp2(towerNONote n) → 0 in $\mathbb{Z}_2$ (ZP-L §IV)
B ↔ C (type bridge)	$c_1 \leftrightarrow 0 \in \mathbb{Z}_2$	snapEmbed $c_1 = 0$ (§I above)

Theorem: snap_state_zp2_is_zero (Incompleteness.lean §III)
snapEmbed ((λ α : Ordinal, if $\alpha < \varepsilon_0$ then c_0 else c_1) ε_0) = 0
The canonical snap map assigns c_1 at ε_0; snapEmbed sends c_1 to 0.
The snap state at the ordinal threshold maps to the 2-adic zero.
Proof: simp (the if-condition evaluates to False at $\alpha = \varepsilon_0$).
Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: zpm_triangle (Incompleteness.lean §III)
All three edges of the triangle, co-proved:
(A ↔ C) $\forall\ n : \mathbb{N}$, fundamentalSeq n $< \varepsilon_0$
(A ↔ C) (λ α, if $\alpha < \varepsilon_0$ then c_0 else c_1) $\varepsilon_0 = c_1$

Theorem: zpm_triangle (Incompleteness.lean §III)
(A ↔ B) Filter.Tendsto (λ n, cnfToZp2 (towerNONote n)) Filter.atTop (nhds 0)
(B ↔ C) snapEmbed (λ α, if α < ε_0 then c_0 else c_1) $\varepsilon_0 = 0$
Proof: ⟨epsilonZero_tower_lt, if_neg (lt_irrefl ε_0), tower_converges_to_zero, snap_state_zp2_is_zero⟩.
Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Section IV: The Kleene and Ordinal Fixed Points

The Kleene recursion theorem (ZP-K) gives a code fixed up to evaluation, and ε_0 (ZP-L) is the least fixed point of ω^·; the two come from different constructions. In computability theory the construction is a diagonal argument over codes and their Gödel numbers. In ordinal theory ε_0 is reached by iterating the normal function α ↦ ω^α from 0 and taking the supremum (ε_0 = nfp (ω^·) 0, epsilonZero_eq_nfp in ZeroParadox/Ordinal/Gentzen.lean), the fixed-point lemma for normal functions rather than a diagonal argument. The two domains share no mathematical machinery. The theorem below records the ordinal fixed point and, on the computability side, the computability fixed point the recursion theorem yields (IsComputationalQuine), a condition constant codes also meet, so it does not pin self-reference (Remark R-M.1).

Domain	Operation	Fixed point
Computability (ZP-K)	eval c = selfApply c (c runs on its own Gödel number)	∃ c : Code, IsComputationalQuine c
Ordinal theory (ZP-L)	ω^α = α (α is a fixed point of ω-exponentiation)	ε_0 = nfp (ω^·) 0, the least such α

Theorem: both_fixed_points_exist (Incompleteness.lean §IV)
(∃ c : Code, ∀ n, eval c n = eval c (encode c + n)) ∧
(∃ α : Ordinal, ω^α = α ∧ ∀ β, ω^β = β → α ≤ β)
Two existence statements, co-proved in one formal context.
Left: a code periodic with period its own Gödel number, via computational_quine_exists; constant codes also satisfy it, so the left conjunct alone does not pin self-reference.
Right: $\varepsilon_0 = \omega^{\varepsilon_0}$, minimal; via epsilonZero_fixedPoint and epsilonZero_le_fixedPoint.
The co-existence is the formal content: two existence statements, two domains, one theorem.
Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Remark R-M.1: DA-1 Path 2 and the Limits of the Diagonalization Frame

both_fixed_points_exist records, on one side, a code periodic with period its own Gödel number (obtained by Kleene diagonalization, though constant codes also meet the condition) and, on the other, the ordinal $\varepsilon_0 = \omega\hat{}\varepsilon_0$, the least fixed point and the supremum of the iterates of $\omega\hat{}\cdot$ from 0. The statement is a conjunction of two existentials, so it establishes that both exist, not that they are one structural pattern; reading them as the same pattern is the framework's interpretation, and no equation between a Code and an Ordinal is well-formed. The Kleene fixed point comes from a diagonal argument (Lawvere 1969; Yanofsky 2003, Theorem 5, places the recursion theorem under that schema); ε_0 comes from the fixed-point lemma for normal functions, by iteration to a supremum.

The informational argument has two charts. Depth chart, ZP-C L-INF: surprisal is unbounded above in the ball-hierarchy depth, each value finite (l_inf). Point chart: at 0 of $\mathbb{Q}_2$, the one point in every ball of the chain centred at 0, the 2-adic additive valuation is $\top$ (padic_addVal_bot), the point to which c_0, $\bot$ of MachinePhase, corresponds in ZP-C. A Lean map carrying that correspondence was not located as of 2026-11-01, searched as follows: the .lean files under ZeroParadox/ for a map whose domain is MachinePhase and whose codomain is $\mathbb{Z}_2$ or $\mathbb{Q}_2$ (only snapEmbed, and a negative example in ZeroParadox/Ordinal/Gentzen.lean), for a definition sending c_0 to 0 (none), and for any declaration naming both MachinePhase and a 2-adic type (those two files only). § I's snapEmbed is a different map, sending c_0 to 1 and c_1 to 0; the two are not composed. L-INF is structurally analogous to the Kleene fixed point. Whether it fits Kleene's diagonalization schema formally is the open question this remark tracks.

L-INF stays formally unconnected. It is a measure-theoretic statement (surprisal under a probability measure on ZP-B's binary space), while the Kleene quine is a computability-theoretic statement (partial recursive functions, Gödel encoding), and neither this remark nor both_fixed_points_exist links them. ZP-C uses the surprisal measure and Kolmogorov complexity as two separate arguments (ZP-C Remark R-BRIDGE).

DA-1 Path 2 is a different matter. In ZP-E (DA-1 insert § IV) it is the step from unbounded surprisal to executing: a bridge principle of its own, a missing principle and not a missing proof, which the framework does not adopt and which is not the occurrence commitment (ZeroParadox/Ordinal/Incompleteness.md, Remark R-M.1).

Theorem Summary

Theorem	Section	Status
snapEmbed_injective	§I	Proved ✓
snapEmbed_mul_morphism	§I	Proved ✓
snapEmbed_c0_val	§I	Proved ✓
snapEmbed_c1_dvd	§I	Proved ✓
hfp_from_epsilon_zero	§II	Proved ✓
snap_unconditional	§II	Proved ✓
snap_state_zp2_is_zero	§III	Proved ✓
zpm_triangle	§III	Proved ✓
both_fixed_points_exist	§IV	Proved ✓

Axiom Purity
All theorems carry axiom footprint: [propext, Classical.choice, Quot.sound].
These are standard Mathlib infrastructure axioms, inherited from ordinal theory (ZP-L), 2-adic analysis (ZP-B), and computability theory (ZP-K).
Classical.choice is carried by the statements of the ordinal fixed-point (nfp) results as written (the statement controls printed in ZeroParadox/Ordinal/Epsilon0MinMax.lean; ZeroParadox/Category/ChoiceCannotBe.md), and present in the computability fixed-point (Kleene's theorem). ZP-K Section IV tabulates the per-declaration measurements.
The footprints listed here are copies; to re-measure one, run #print axioms on the named declaration.

End of ZP-M | Kleene–Ordinal Bridge | snapEmbed type bridge | hfp gap closed | zpm_triangle co-proved | both_fixed_points_exist | R-M.1: DA-1 Path 2 boundary | All Incompleteness.lean theorems verified. Axioms: [propext, Classical.choice, Quot.sound].